Mathematica Basics

Computer Graphics, Wolfram Mathematica

In[1]:=
```
SetOptions[{Plot, Plot3D, DensityPlot, ParametricPlot,
    ParametricPlot3D, ListPlot, Graphics, Graphics3D, SphericalPlot3D, DensityPlot,
    DensityPlot3D, ContourPlot, SliceContourPlot3D, SliceDensityPlot3D,
    RegionPlot, RegionPlot3D, Graphics3D, RevolutionPlot3D}, ImageSize → Small];
```

In[2]:=
```
SetOptions[EvaluationNotebook[], ShowCellLabel → True, LineSpacing → {1.5, 0}];
```

In[3]:=
```
(* === Chapter 1 === *)
```

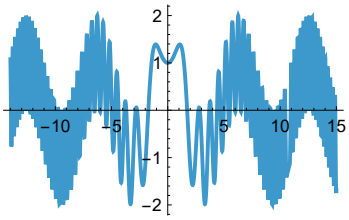
In[4]:=
```
{15. / 40, 15 / 40, Cos[Pi / 6.], Cos[Pi / 6]}
```

Out[4]=
$$\left\{0.375, \frac{3}{8}, 0.866025, \frac{\sqrt{3}}{2}\right\}$$

In[5]:=
```
f11[x_] := Sin[x^2] + Cos[x]
```

In[6]:=
```
Plot[f11[x], {x, -14, 15}]
```

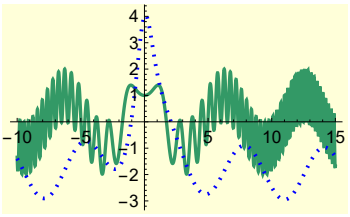
Out[6]=



In[7]:=
```
f12[x_] := Sin[x] + 6 / (1 + x^2) - 2
```

In[8]:=
```
Plot[{f11[x], f12[x]}, {x, -10, 15},
 PlotStyle → {{RGBColor[0.2, 0.6, 0.4], Thickness[0.01]}, {Blue, Thick, Dotted}},
 Background → LightYellow]
```

Out[8]=



In[9]:=
```
M1 = {{1, 2, 3}, {4, 5, 6}}
```

Out[9]=
```
{{1, 2, 3}, {4, 5, 6}}
```

In[10]:=
```
{MatrixForm[M1], M1[[2]], M1[[2]][[1]]}
```

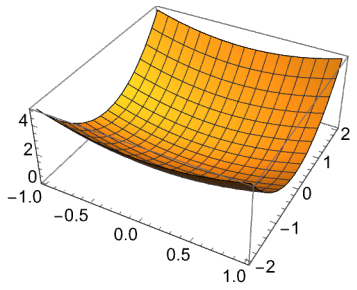
Out[10]=

$$\left\{\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \{4, 5, 6\}, 4\right\}$$

In[11]:=
```
f13[x_, y_] := x^2 + y^2
```

In[12]:=
```
Plot3D[f13[x, y], {x, -1, 1}, {y, -2, 2}]
```

Out[12]=



In[13]:=
```
{f13[1 + a, Sqrt[1 + a + a^2]], Simplify[f13[1 + a, Sqrt[1 + a + a^2]]]}
```

Out[13]=

$$\left\{1 + a + a^2 + (1 + a)^2,\ 2 + 3\,a + 2\,a^2\right\}$$

In[14]:=
```
fx11[t_] := t + 3
fy11[t_] := 2 t + 3
ft11[t_] := {fx11[t], fy11[t]}
```

In[17]:=
```
ParametricPlot[ft11[t], {t, -5, 5}, AspectRatio → 1]
```

Out[17]=

